The Earth spins relative to what?

Geodetic Reference Systems — Part 2: the celestial reference systems (ICRS and ICRF) and the bridge to the ground — four classes

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Astrometry and Coordinate Systems Simulators

What we will see

1. Where we left off
2. Two surprises
3. Why a sky that does not spin?
4. The sky as a sphere
5. The sky moves
6. Truly still
7. VLBI: listening with two ears
8. From the ICRS to the ICRF
9. Hipparcos, Gaia and the Galaxy's acceleration
10. Two views, one rotation
11. The Earth's rotation
12. The axis wobbles in space
13. The pole moves inside the Earth
14. The EOP: who measures and who publishes
15. Several clocks, one hour
16. Satellite time
17. From the orbit to the receiver
18. Brazil and the wrap-up

This is Part 2

Part 1 (the previous class)

The **terrestrial** reference systems: ITRS and ITRF, SIRGAS2000, WGS 84. The ones that *turn with the Earth*.

Part 2 (this class)

The **celestial** reference systems: ICRS and ICRF. The ones that **do not turn with the Earth** — and the **bridge** that links the sky to the ground.

Part 2 also **starts from scratch**: if you already know what right ascension is, it goes quickly — but its ideas all come back in class 3.

Four classes

Class 1 · A sky that does not spin

Why a reference system that does not turn with the Earth; coordinates of a star; what, in the sky, *moves*.

Class 2 · The beacons of the Universe

Quasars, VLBI, the ICRS and the ICRF; then Hipparcos and Gaia.

Class 3 · The sky–ground bridge

How the Earth is turned in space: rotation, precession, nutation, pole — and the EOP.

Class 4 · Time and practice

UT1, UTC, TAI, the leap second and GNSS time; from the orbit to the receiver; Brazil.

Each class is 50 minutes long, starts by recalling the previous one and ends with a summary. We will go **slowly**: one idea per slide.

Part 1

A sky that does not spin

class 1 of 4

Where we left off

Where we left off: Part 1

In Part 1, on the **terrestrial** reference systems:

1. a coordinate must say in which **frame** and at which **epoch**;
2. the **system** is the written rule; the **frame**, the real stations;
3. the ITRS/ITRF is **fixed to the Earth**: it turns with it;
4. its orientation comes from a 1984 agreement — and what ties it down are **quasars**, seen by VLBI.

One piece is missing

All of that serves those who are at rest on the Earth. What about those who are not — a satellite, the Moon, a star, a quasar?

They need a reference system that **does not turn with the Earth**.

Today's questions

1. Why do we need a reference system that does not turn with the Earth?
2. How does a star get coordinates?
3. What does "J2000.0" mean in the catalog?
4. The sky looks still — but what, in fact, *moves*?

Deal

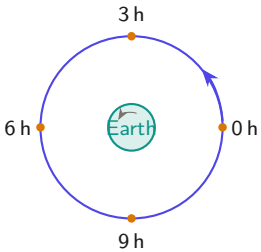
By the end of the class you will answer all four — in your own words.

They come back on the last slide.

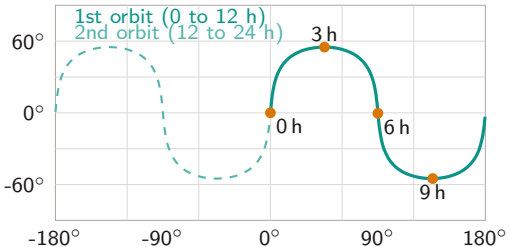
Two surprises

The same satellite, two drawings

A GPS satellite goes once around the Earth in about **12 hours**. The same satellite, over one day, in two drawings:



seen from space: a circle



seen from a map fixed to the Earth: a twisted loop

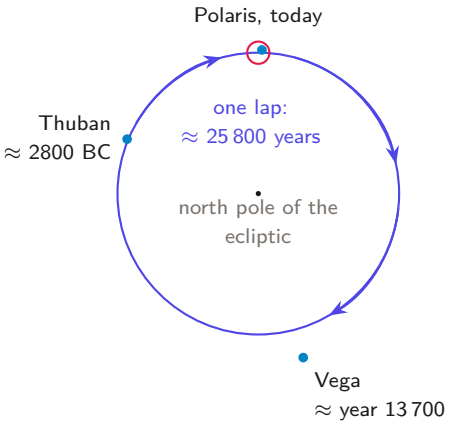
The satellite is the same. The **reference system** changed: on the right, the map turns underneath it (one turn every 23 h 56 min).

The north of the sky changes address

Today the north celestial pole points almost at **Polaris**, the North Star: it lies less than 1° away.

But the Earth's axis does not always point to the same place. About 4 800 years ago, in the time of the pyramids, the "north star" was **Thuban**. In about 11 700 years it will be **Vega**.

In the Southern Hemisphere there is no bright star at the pole: the Southern Cross serves as the guide.



The celestial pole walks along a circle of radius 23.4°.

The thesis of Part 2

Two surprises, one lesson: what is "still" **depends on who is looking**.

Who needs a reference system that does not turn with the Earth?

- **satellites** (GNSS, laser, DORIS): the orbit obeys gravity in space, not the map;
- **probes** and missions to the Moon and Mars;
- **astronomy**: stars, planets, galaxies;
- the **Earth itself**: to measure how much it turns, you need something that does not turn with it.

The question that organizes the four classes

Spinning **relative to what**?

Where we are headed

The **ICRS** system and the **ICRF** frame: a sky tied to **quasars**, so distant that they look motionless.

Why a sky that does not spin?

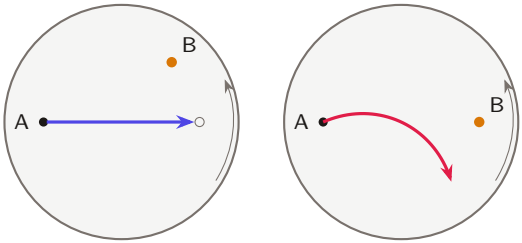
The carousel, again

You are on a carousel and throw a ball to a friend on the other side.

Seen from outside, the ball goes in a straight line — but your friend turned and "got out of the way".

Seen from the carousel, the ball makes a curve, as if someone pushed it.

Remember the "little push outward" from the gravimetry class? It only appears because we are in a reference system that turns.



seen from outside: straight

seen from the carousel: curved

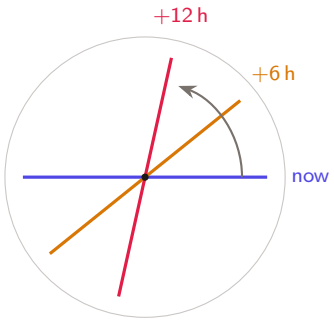
The Earth leaves clues, but does not say “relative to what”

On a closed carousel you can still *feel* that you are turning. The Earth is similar: in 1851, in Paris, Foucault hung a **67-meter** pendulum in the Panthéon — and its plane of swing **turned** by itself.

In **Curitiba**: one turn every **2.3 days** (about 155° per day), counterclockwise seen from above.

But an answer is missing

The pendulum proves that the Earth turns, but does not say relative to what. Newton imagined an “absolute space”; **in practice, we use the most distant objects we can see**.



seen from above, in Curitiba

The sky as a sphere

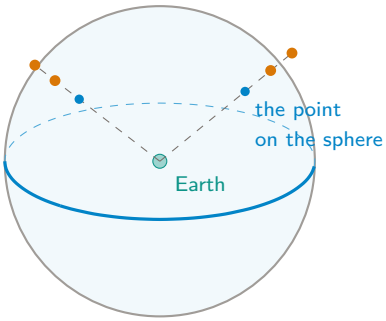
The celestial sphere

Stars are at very different distances. But to say *where* a star is in the sky, only the **direction** matters.

So we imagine a giant sphere around the Earth, like the dome of a planetarium, and "paint" each object at the point where the line of sight crosses it: the **celestial sphere**.

Direction, not position

In Part 1, a coordinate was a position, in meters. Here, it is only a direction: **two angles**.



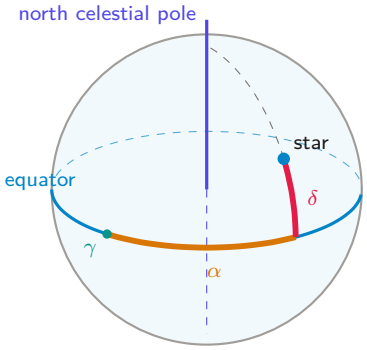
stars at different distances,
in the same direction

Latitude and longitude of the sky

On the Earth, we give latitude and longitude. In the sky, it is the same idea:

- the **celestial equator** is the Earth's equator "stretched" out to the sphere; the **celestial poles** are the Earth's axis stretched out;
- **declination** δ: the "latitude" — from $-90°$ to $+90°$;
- **right ascension** α: the "longitude" — measured eastward, from a starting point γ.

Instead of degrees, right ascension usually comes in **hours**.



Hours become degrees

The sky makes one turn in 24 hours.

So:

$$\begin{aligned} 24\text{ h} &= 360^\circ \\ 1\text{ h} &= 15^\circ \\ 1\text{ min} &= 15' \\ 1\text{ s} &= 15'' \end{aligned}$$

(minutes and seconds *of time* become minutes and seconds *of arc*, times 15.)

	α	δ
Sirius	6 h 45 min 8.9 s	$-16^\circ 42' 58''$
	$= 101.287^\circ$	$= -16.716^\circ$
Acrux	12 h 26 min 35.9 s	$-63^\circ 05' 57''$
	$= 186.650^\circ$	$= -63.099^\circ$

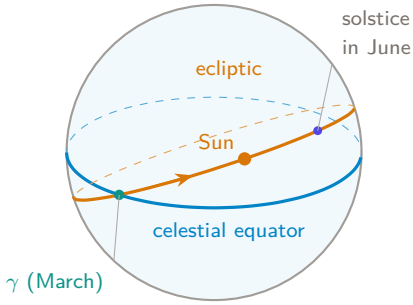
Sirius is the brightest star in the night sky; Acrux is the "foot" star of the **Southern Cross**. Values for the equinox and epoch J2000.0.

The starting point: ecliptic and vernal point

For longitude on the Earth, the world agreed on the Greenwich meridian. And for right ascension?

Over the year, the Sun walks among the stars along a line: the **ecliptic**. It is tilted by 23.4° relative to the celestial equator — the same tilt as the Earth's axis, the cause of the seasons.

The two lines cross at two points, the **equinoxes**. The March one is the **vernal point** γ: where the Sun crosses the equator, from south to north. It is the zero of right ascension.



The angle between the two lines is 23.4°.

Your turn

Three little sums

(a) Vega has right ascension 18 h 36 min 57 s. How many degrees?

(b) A star has $\alpha = 150^\circ$ and $\delta = -30^\circ$. What is the right ascension in hours? Is it in the northern or southern sky?

(c) How many degrees does the Earth turn in **4 minutes**?

Your turn

Three little sums

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Answers

(a) $18.6158\ \text{h} \times 15 = 279.24^\circ$.

(b) $150/15 = \mathbf{10}$ h, **southern** sky (negative declination).

(c) $4 \times 15' = 60' = \mathbf{1}^\circ$. Remember: 4 minutes of time = 1 degree.

The sky moves

What “J2000.0” means

Every star catalog carries, next to the coordinates, a label: **equinox and epoch J2000.0**.

J2000.0 is **1 January 2000, at noon**.

It is the same lesson as in Part 1: a coordinate must say in which **frame** and at which **epoch**. In the sky, the label says: “the starting points — the equator and the equinox — are the ones that existed at that instant”.

Why the label is needed

Because the two starting points move, and the stars themselves too.

Three things move

1. the equinox and the equator: **precession**;
2. the equator again, rippling: **nutation**;
3. the stars: **proper motion**.

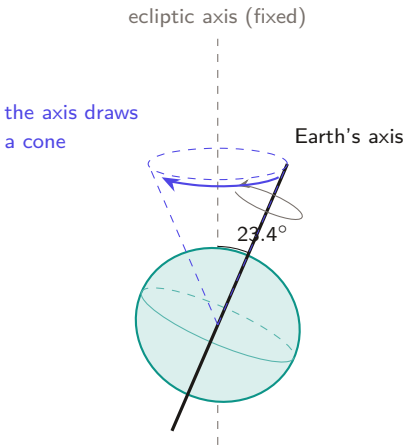
Precession: the Earth's spinning top

Spin a top while it is tilted. Besides spinning around its own axis, its axis slowly draws a **cone**.

The Earth does the same: its axis turns around the axis of the ecliptic, keeping the tilt of 23.4°. One full turn takes ≈ 25 800 **years**.

The name

This slow motion of the axis is **precession**. It is what moved the celestial pole from Thuban to Polaris, and will take it to Vega.

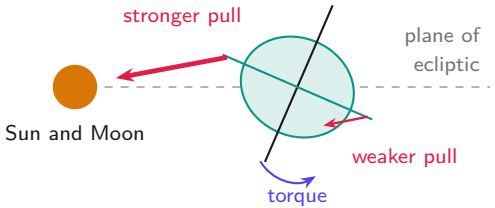


What pushes the top?

The Earth is not a perfect sphere: its radius at the Equator is **21 km larger** than at the poles — a "belly".

The Sun and the Moon pull harder on the side of the belly that is *nearer*, and less on the far side. The difference is a torque that tries to straighten the axis.

But the Earth spins, like a gyroscope. Instead of toppling, the axis **goes around**: that is precession.



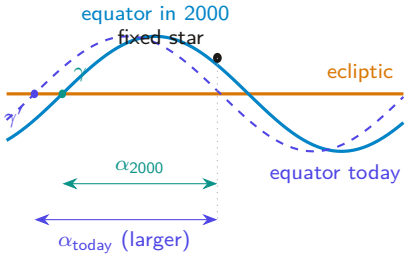
The vernal point slips

If the celestial equator changes place, the point where it crosses the ecliptic — the vernal point — **slips westward**, along the ecliptic:

$50.29''$ per year

That is 1.4° per century, or 1° every **72 years**.

And since right ascension is measured *from the vernal point*, the right ascension of every "fixed" star keeps changing.



(shift exaggerated)

Stop and think

The vernal point slips **westward**. Right ascension is measured **eastward**, from it.

The question

Is the right ascension of a fixed star, today, **larger** or **smaller** than in 2000? And by how much, for Sirius?

Stop and think

The vernal point slips **westward**. Right ascension is measured **eastward**, from it.

The question

Is the right ascension of a fixed star, today, **larger** or **smaller** than in 2000? And by how much, for Sirius?

Answer

Larger. The starting point moved backward, so the distance measured from it, forward, *grows*.

Sirius: from 101.287° to 101.587° — about $+1\,078''$ ($+0.30^\circ$) in 26.8 years.

Hipparchus and Spica (about 129 BC)

The Greek astronomer **Hipparchus** compared the position of **Spica** with observations made about 150 years earlier by Timocharis.

Spica was about 2° farther from the vernal point. The stars had not moved that much: what had moved was the starting point.

Hipparchus estimated about 1° per century (today's value is 1.4°). It was the first time anyone realized that the sky has an **epoch**.

The sum

144 years $\times$ $50.29''$ per year $=$ $7\,242'' \approx \mathbf{2.0^\circ}$.

It checks out: the shift measured by the Greeks is what today's precession predicts.

Example: Sirius in J2000.0 and today

	α	δ
J2000.0	101.2872°	-16.7161°
today (2026-10-18)	101.5865°	-16.7457°
difference	$+1\,078''$	$-106''$

Precession only (mean equinox and equator of date). Sirius's proper motion adds $\approx -33''$ in δ.

How big that is

Total: $0.288^\circ = 17.3'$.

That is **60 % of the diameter of the full Moon**.

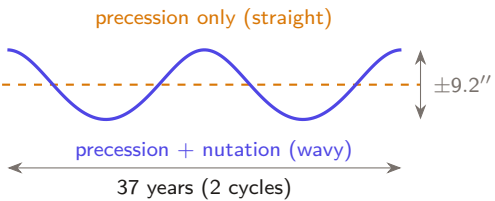
In 27 years, the "same" catalog point stopped being where the star is.

Nutation: the top wobbles

The Moon complicates things. Its orbit is tilted about 5° relative to the ecliptic, and the plane of that orbit also turns: one turn in **18.6 years**.

So the Moon's pull on the belly varies, and the Earth's axis "wobbles" while it goes around its circle: **nutation**.

Amplitudes: up to $17.2''$ in longitude and $9.2''$ in the tilt.

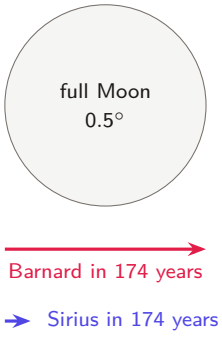


Path of the pole in the sky. Vertical scale enlarged ≈ 7 times.

Stars move too

Stars travel through the Galaxy at tens of kilometers per second. For the closest ones, the apparent shift can be measured: the **proper motion**.

- **Barnard's Star**: $10.4''$ per year — the record. In **174 years** it moves the diameter of the Moon.
- **Sirius**: $1.34''$ per year.
- Most stars: far less than $0.1''$ per year.



What moves, in one table

what moves	how much	what changes in the coordinates
the equinox (precession)	$50.29''/\text{yr}$	the right ascension of all stars
the pole (precession)	$20''/\text{yr}$	the declination of all stars
the equator (nutation)	$\pm 17''$ in 18.6 yr	a ripple in both
the stars (proper motion)	up to $10''/\text{yr}$	each star, in its own way

Nothing we have used so far as "still" is really still. Now what?

Try it: the Astrometry Simulator

In the `astrometria` folder you are a surveyor without GNSS: you point the theodolite at three stars, record the *zenith distance* and the *horizontal angle*, and the program adjusts, by **least squares**, the latitude, the longitude and the azimuth constant.

- catalog of $\approx 2\,800$ stars (RA and Dec in degrees, J2000.0);
- "Greenwich Mean Time" becomes sidereal time; local sidereal time = Greenwich + longitude.

What the simulator leaves out

- **precession**: coordinates are those of J2000.0 — in 2026, up to 0.37° off (≈ 40 km);
- **nutation**: up to $17''$ (≈ 0.5 km);
- **UT1**$-$**UTC**: the time is treated as UT1; the error reaches 0.9 s $= 13.5''$ (≈ 0.4 km at the Equator).

These are good simplifications for teaching. Organizing what is left out is the subject of this Part 2.

Take-home — class 1

1. Whatever does not turn with the Earth — satellites, probes, stars — needs a reference system that **does not turn either**. "Turning" is always *relative to something*.
2. The sky is a **sphere** of directions. Coordinates are two angles: **right ascension** α (from γ, eastward; 24 h $= 360^\circ$) and **declination** δ.
3. γ, the vernal point, is where the ecliptic crosses the celestial equator. The catalog says "**J2000.0**": *frame and epoch*, as in Part 1.
4. **Precession**: the Earth's axis makes one turn in $\approx 25\,800$ years; the vernal point slips $50.29''$/yr. **Nutation**: $\pm 17''$ in 18.6 years. **Proper motion**: stars move.
5. So "fixed" exists neither in the equator nor in the stars. We need something **even more distant and motionless**.

Next class: the beacons of the Universe — quasars, VLBI and the **ICRS** / **ICRF**. **Try it:** the Astrometry Simulator, adjusting your position with three stars.

Part 2

The beacons of the Universe

class 2 of 4

Truly still

Where we left off: class 1

In class 1:

1. the sky is a **sphere of directions**, with two coordinates: right ascension α and declination δ;
2. they depend on the equator and the equinox — and those **move**: precession ($50.29''$/yr) and nutation ($\pm 17''$);
3. the stars themselves move (up to $10''$/yr).

Today's question

What stays still for real?

Provisional answer: something so far away that, to us, it does not move.

Today's questions

1. What stays still for real?
2. How do we measure the direction of something so distant, to within millionths of an arcsecond?
3. What are the **ICRS** and the **ICRF**: the system and the frame of the sky?
4. Where do Hipparcos and Gaia come in?

Deal

By the end of the class you will answer all four — in your own words.

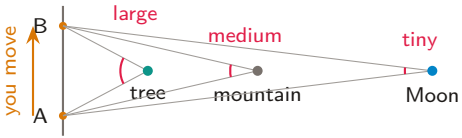
The farther away, the less it moves

On a car trip, the trees by the roadside fly past. The mountain far away moves slowly. And the Moon seems to follow you, almost motionless.

For the same motion of yours, the apparent shift is **inversely proportional to the distance**: twice the distance, half the shift.

The idea

Want something that does not move? Look for the thing that is **farthest away** you can see.



The angle the object "sees" (in red) shrinks with distance.

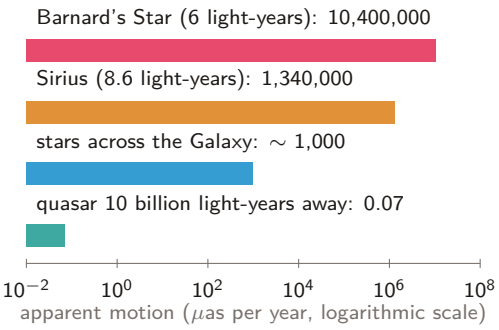
How big is the apparent shift?

Something moving sideways, at speed v and distance d, seems to move across the sky at

$$\boxed{\frac{v}{d}}$$

radians per second.

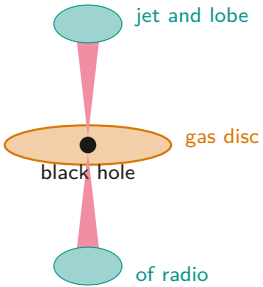
Barnard moves **150 million times more** than a quasar.



What is a quasar?

At the center of some galaxies sits a **giant black hole** (millions to billions of solar masses) swallowing gas. The gas heats up as it falls and outshines the whole galaxy; jets of particles leave through the poles.

From afar it is just a point of light — "almost a star": *quasi-stellar radio source*, or **quasar**. The first with a deciphered spectrum was **3C 273** (M. Schmidt, 1963): its light left about **2 billion years ago**.



sketch; not to scale

Why radio?

Many quasars are strong **radio** emitters; radio waves cross clouds, work day and night — and, as we will see, distant antennas can be combined.

VLBI: listening with two ears

How sharp must the gaze be?

A single antenna only sees details larger than roughly the wavelength divided by the diameter:

$$\theta \approx 1.22 \, \frac{\lambda}{D}$$

- human eye: $\approx 60''$;
- **100 m** dish, 3.6 cm waves: $\approx 90''$ — same as the eye;
- two antennas **10 000 km** apart: $\lambda/B \approx$ **0.7** mas.

That is **120 000 times sharper**. And we still want to reach 30 μas.

The idea of VLBI

Very Long Baseline Interferometry: instead of one giant antenna, **two very distant antennas** — a telescope the size of the Earth, made of pieces.

Fun fact

The same technique, with antennas on several continents, gave in 2019 the first **image of a black hole**: the one in galaxy M87.

Two ears and a thunderclap

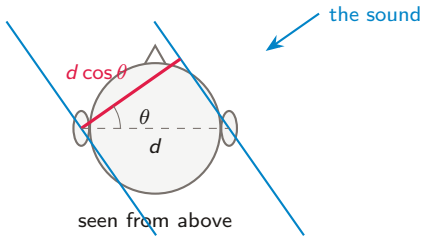
You know which side a thunderclap comes from: the sound reaches one ear a little *before* the other. The brain measures that **delay** and gives the direction.

Ears $d = 0.20$ m apart, sound at 343 m/s: the maximum delay is

$$\frac{d}{v} = \frac{0.20}{343} \approx 0.6 \text{ ms}$$

And the delay depends on the angle θ:

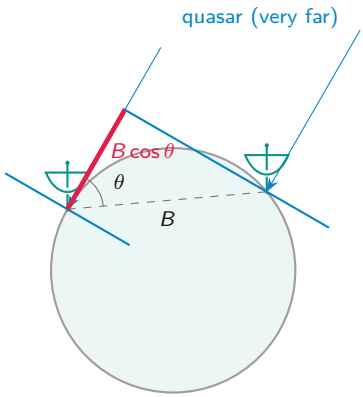
$$\boxed{\text{delay} = \frac{d\,\cos\theta}{v}}$$



The same, with radio waves

Replace the ears by **two distant antennas** and the sound by radio waves from a quasar. The wave reaches one antenna before the other; measure the delay τ and θ follows:

$$\tau = \frac{B \cos\theta}{c}$$



- B: the **baseline**, the distance between the antennas;
- θ: the angle between the baseline and the quasar direction;
- c: the speed of light.

From the delay to the angle

Inverting the formula:

$$\cos\theta = \frac{c\,\tau}{B}, \qquad \delta\theta \approx \frac{c\,\delta\tau}{B\,\sin\theta}$$

With a baseline of $B = 10\,000$ km:	
maximum delay B/c	33.4 ms
precision of the delay $\delta\tau$	10 ps
(light in 10 ps travels)	3 mm
precision of the angle $\delta\theta$	$\approx 60\,\mu$as

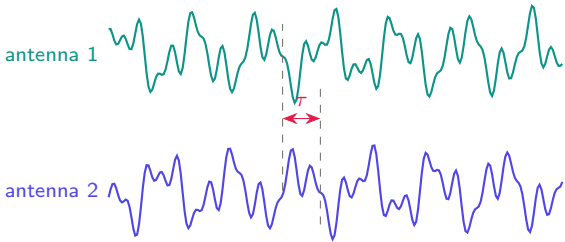
How small is 10 ps?

10 ps $= 0.000\,000\,000\,01$ s: one hundredth of a billionth of a second. To measure the delay that well, you need an extraordinary clock at each antenna.

How to record without wires

The antennas are thousands of kilometers apart: you cannot run a wire between them.

1. Each antenna **records** the radio "hiss" of the quasar, together with the time from an **atomic clock** (a hydrogen maser).
2. The recordings go to a computer, the **correlator**.
3. It **slides** one recording over the other until the two match. The slide is the delay τ.



the same signal, recorded in two places

Your turn

A 6 000 km baseline

(a) A quasar is at $\theta = 60°$ from the baseline. What is the delay τ?

(b) To know θ to $1\,\mathrm{mas}$, how precisely must we measure τ?

(c) How far does light travel in that time?

Your turn

A 6 000 km baseline

(a) A quasar is at $\theta = 60°$ from the baseline. What is the delay τ?

(b) To know θ to $1\,\mathrm{mas}$, how precisely must we measure τ?

(c) How far does light travel in that time?

Answers

(a) $\tau = \dfrac{6 \times 10^6 \cos 60°}{299\,792\,458} \approx 10.0$ ms.

(b) $\delta\tau = \dfrac{B \sin\theta\, \delta\theta}{c} \approx \mathbf{84}$ ps
($1\,\mathrm{mas} = 4.85 \times 10^{-9}$ rad).

(c) $c \times 84$ ps ≈ 2.5 cm.

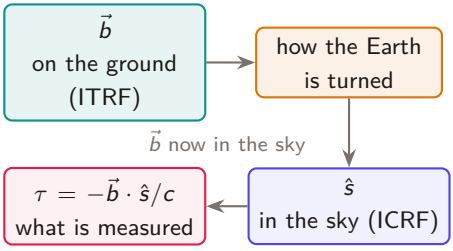
The delay knows both reference systems

The delay, written with vectors:

$$\tau = -\frac{\vec{b} \cdot \hat{s}}{c}$$

- $\vec{b}$: the baseline — known in the **ITRF**, fixed to the Earth;
- $\hat{s}$: the direction of the quasar — in the **ICRF**, fixed to the sky.

To multiply, both must be **in the same reference system**: you need to know *how the Earth is turned*. That is class 3.

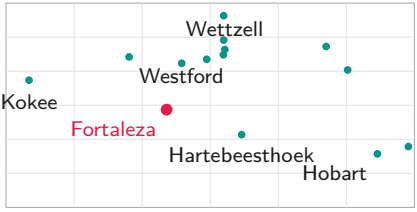


Many delays, few unknowns: **least squares**, again. The unknowns: antennas, quasars, Earth orientation, clocks, atmosphere.

The IVS and the antennas around the world

The **IVS** (*International VLBI Service for Geodesy and Astrometry*) combines the observatories of dozens of countries.

- **24 h** sessions, a few per week: the full orientation of the Earth;
- **1 h** sessions ("*Intensives*"), almost every day: the rotation (UT1−UTC), with fast results;
- the new generation, **VGOS**: 12 m antennas and a goal of **1 mm**.



some stations of the network (approximate positions)

In Brazil

Fortaleza, Northeast Space Radio Observatory (Eusébio, CE): 14.2 m antenna in VLBI since **1993**; in November 2025 a new antenna with a **VGOS** terminal was inaugurated. The Southern Hemisphere has few antennas: each one counts a lot.

From the ICRS to the ICRF

System and frame, again

	terrestrial (Part 1)	celestial (Part 2)
system	ITRS	**ICRS**
frame	ITRF2020	**ICRF3**
what the list holds	stations: coordinate *and* velocity	quasars: right ascension and declination
turns with the Earth?	yes	**no**
who adopts it	IUGG / IERS	IAU (with the IERS)

The same pair: the **written rule** and the **real things**.

The rule of the ICRS, in four lines

origin	the **barycenter** (center of mass) of the Solar System
orientation	axes **tied to distant extragalactic objects**, chosen to coincide with the mean equator and equinox of J2000.0 (to $\approx 18\,\mathrm{mas}$), so as not to "break" the old catalogs
scale	directions only; the surrounding space-time uses the SI meter and second
time	the axes **do not rotate** relative to the distant Universe

Notice what is missing here

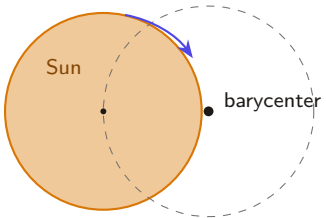
No source, no coordinate: nothing you can measure. This is the system — and that is why it is not enough.

Why the barycenter, and not the Earth?

The Sun does not stay put: it and the planets turn around the **center of mass** of the whole, the barycenter.

Jupiter alone already pulls the barycenter *outside* the Sun's surface: 1.07 solar radii from its center.

And the Earth accelerates as it goes around the Sun: a reference system at its center is not "still". The barycenter is the best resting point the Solar System offers.



the Sun's center circles around the barycenter

The frame: the ICRF

The **ICRF** is a **list of positions**: for each quasar, the name, the right ascension and the declination — with the uncertainty.

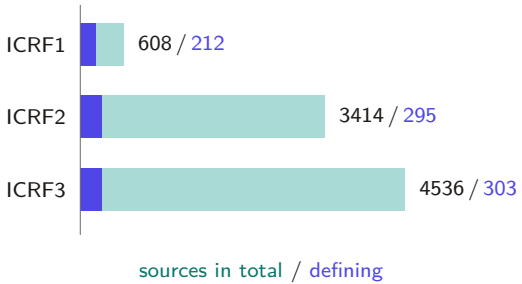
Since the list is measured *with VLBI*, the precision is thousandths of an arcsecond and better.

Same form as the ITRF: names and numbers. Only the numbers are **directions**, and they do not age with precession.

source	α	δ	what it is
3C 273	$12^{\mathrm{h}}\,29^{\mathrm{m}}\,6.70^{\mathrm{s}}$	$+2^\circ\,3'\,8.6''$	quasar (1963)
3C 279	$12^{\mathrm{h}}\,56^{\mathrm{m}}\,11.17^{\mathrm{s}}$	$-5^\circ\,47'\,21.5''$	variable quasar
3C 84	$3^{\mathrm{h}}\,19^{\mathrm{m}}\,48.16^{\mathrm{s}}$	$+41^\circ\,30'\,42.1''$	galaxy NGC 1275
3C 345	$16^{\mathrm{h}}\,42^{\mathrm{m}}\,58.81^{\mathrm{s}}$	$+39^\circ\,48'\,37.0''$	quasar with a jet
M87	$12^{\mathrm{h}}\,30^{\mathrm{m}}\,49.42^{\mathrm{s}}$	$+12^\circ\,23'\,28.0''$	giant galaxy

Famous sources; rounded catalog values (J2000.0). "3C" = Third Cambridge Catalogue of radio sources.

Three generations of the ICRF



	adopted (IAU)	in force
ICRF1	1997 (Kyoto)	1/1/1998
ICRF2	2009 (Rio de Janeiro)	1/1/2010
ICRF3	2018 (Vienna)	1/1/2019

ICRF2 was adopted at the IAU General Assembly **in Rio de Janeiro**. ICRF3 has three frequencies: 8.4 GHz (4,536 sources), 24 GHz (824) and 32 GHz (678).

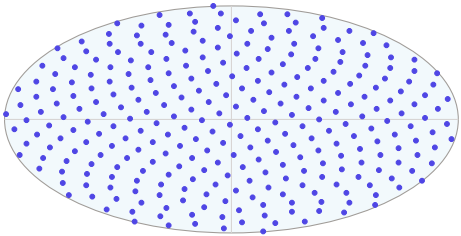
The defining sources

Not every source on the list controls the orientation of the axes. The **defining** ones are chosen (in the ICRF3, **303** out of 4,536): stable position, compact structure and **well spread** over the sky.

It is like choosing the best monuments of a network: the others stay on the list, but do not tie the axes.

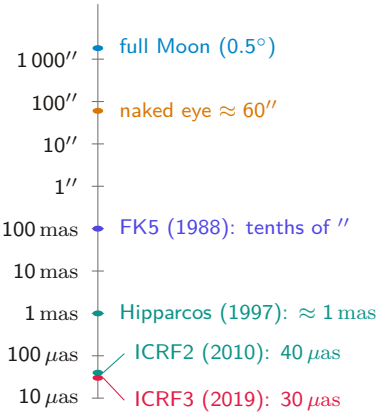
Why only the stable ones?

A quasar has a *jet*: the brightest radio spot can wander a little when the jet changes. That would shake the axes.



303 defining sources (illustrative distribution)

The ladder of angles



What that is worth on the ground

$1''$	$= 30.9$ m
$1\,\mathrm{mas}$	$= 3.1$ cm
$30\,\mu\mathrm{as}$	$= 0.9$ mm

If the reference sky erred by $30\,\mu\mathrm{as}$ in orientation, every point on the Earth would be displaced by about **1 mm**. That is why the 1 mm goal for the ITRF demands a sky of this quality.

To get a feel: $30\,\mu\mathrm{as}$ is the size of about 6 cm *on the Moon*, seen from the Earth.

Your turn

Three little sums

(a) How many centimeters, on the Earth's surface, is $1\,\mathrm{mas}$ worth? (use $1'' = 30.9$ m)

(b) An error of $0.3\,\mathrm{mas}$ in the orientation of the sky displaces points on the Earth by how much?

(c) A quasar moves sideways at 1 000 km/s, 10 billion light-years away. How many μas per year? (Hint: v/d, in radians per year.)

Your turn

Three little sums

(a) How many centimeters, on the Earth's surface, is $1\,\mathrm{mas}$ worth? (use $1'' = 30.9$ m)

(b) An error of $0.3\,\mathrm{mas}$ in the orientation of the sky displaces points on the Earth by how much?

(c) A quasar moves sideways at 1 000 km/s, 10 billion light-years away. How many μas per year? (Hint: v/d, in radians per year.)

Answers

(a) $0.001'' \times 30.9$ m ≈ 3.1 cm.

(b) 0.3×3.1 cm ≈ 0.9 cm (almost 1 cm!).

(c) $v/d = \dfrac{10^6}{10^{10} \times 9.46 \times 10^{15}} \times 3.16 \times 10^7$ $\approx 3.3 \times 10^{-13}$ rad/yr $\approx 0.07\,\mu\mathrm{as/yr}$.

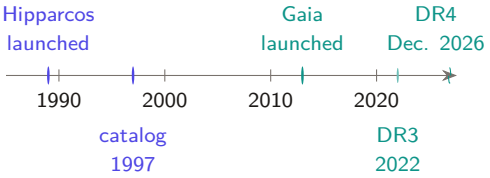
Hipparcos, Gaia and the Galaxy's acceleration

The sky in visible light: Hipparcos and Gaia

Quasars are faint in visible light. For the sky we see, the European Space Agency (ESA) launched two satellites:

- **Hipparcos** (1989–1993): a catalog of 118 218 stars, with $\approx 1\,\mathrm{mas}$ precision (1997);
- **Gaia** (launched 2013; observed until 15 Jan 2025): about **1.8 billion** sources in DR3 (2022), among them $\approx$ **1.6 million quasars**.

Gaia's quasars are fainter and less precise one by one — but there are **millions**.



The next data release, DR4, is planned for 2 Dec 2026.

One system, several frames

frame	light	sources	adopted by the IAU
ICRF3	radio (VLBI)	4,536 (303 defining)	2018 (Res. B2)
Gaia-CRF3	visible	$\approx$ 1.6 million quasars	2021 (Res. B3)
HCRF (Hipparcos)	visible	118,218 stars	1997 (Res. B2)

The **ICRS** is one. The **frames** are several, each in a "color" of light. They are aligned through the sources they share: **3,181** ICRF3 sources are also in Gaia-CRF3.

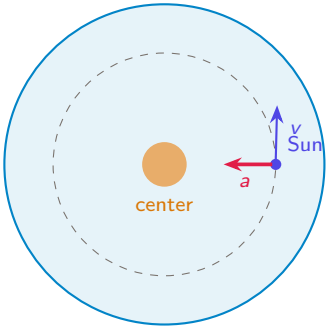
Since 1 Jan 2022 the ICRS has been realized by the ICRF3 (radio) *and* the Gaia-CRF3 (visible). Gaia-CRF3 has largely superseded Hipparcos.

Not even quasars are still: we accelerate

The Solar System goes around the center of the Galaxy at about **230 km/s**: one turn every $\approx$ **220 million years**.

An orbit is an **acceleration** toward the center: $\approx 2.7 \times 10^{-10}$ m/s^2. Effect: all quasars seem to drift toward the center of the Galaxy, by $5.8\,\mu\text{as}$**/yr**.

In 10 years that is $58\,\mu\text{as}$ — more than the ICRF3 noise floor ($30\,\mu\text{as}$). That is why the ICRF3 **corrects this effect**: the positions are valid for epoch **2015.0**.



seen from above, not to scale

Take-home — class 2

1. To get a "still" sky, we use objects **so distant that they do not move**: **quasars** (apparent motion $\approx 0.07\,\mu\mathrm{as/yr}$, against $10''/\mathrm{yr}$ for Barnard's Star).
2. **VLBI**: two distant antennas record the same signal; the delay $\tau = B\cos\theta/c$ gives the angle. With $\delta\tau \approx 10$ ps, $\approx 60\,\mu\mathrm{as}$.
3. **ICRS** (system): origin at the barycenter, axes tied to extragalactic objects, no rotation. **ICRF** (frame): the list of $\approx$ 4,500 quasars (303 defining) — ICRF3, in force since 2019.
4. $30\,\mu\mathrm{as}$ in the sky are worth $\approx$ **1 mm** on the ground.
5. Hipparcos (1997) and Gaia (2013–2025) take care of visible light: **one system, several frames**. And even quasars "move": the Galaxy's acceleration, $5.8\,\mu\mathrm{as/yr}$, is already corrected in the ICRF3.

Next class: how the Earth is turned relative to that sky — rotation, precession, nutation, polar motion — and the **EOP**.

Part 2

The bridge between the sky and the ground

class 3 of 4

Two views, one rotation

Where we left off: class 2

In class 2:

1. **quasars** are so far away that they do not move: they are the still reference;
2. **VLBI** measures the delay $\tau = B\cos\theta/c$ between two antennas and gives the direction to $\approx 60\,\mu\text{as}$;
3. the **ICRS** (system) and the **ICRF** (frame): origin at the barycenter, axes tied to quasars, no rotation.

Today's question

The sky is still. The Earth is turned — but turned by how much, right now?

Provisional answer: by **five numbers per day**.

Today's questions

1. What are the Earth's "turns"?
2. How do we write a turn with numbers?
3. How big is each one, in the sky and on the ground?
4. Who measures, and who publishes? (the **EOP** and the **IERS**)

Deal

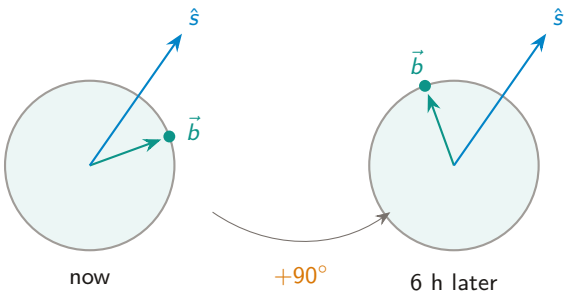
By the end of the class you will answer all four — in your own words.

The same point, two addresses

An antenna sits on the ground: in the **ITRF** its coordinates **do not change**. But, to the sky, it never stops: it makes one turn per day.

The quasar is still in the sky: its direction $\hat{s}$ holds in the **ICRF**.

For the delay $\tau = -\vec{b} \cdot \hat{s}/c$, both vectors must be **at the same address**, that is, in the same reference system. Converting from one to the other is the job of the **bridge**.



Seen from above. $\vec{b}$ turns with the Earth; $\hat{s}$ stays. The angle between them changes, and so does the delay.

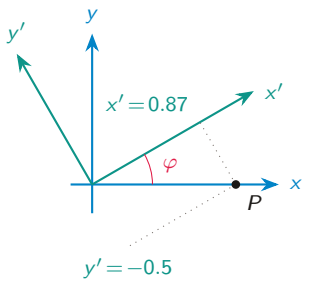
Turning the axes

A point P has coordinates (x, y) in the sky axes. Turn the axes by an angle φ: the point did not move, but the **coordinates change**:

$$x' = x\cos\varphi + y\sin\varphi$$
$$y' = -x\sin\varphi + y\cos\varphi$$

As a table (a **matrix**):

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$



sky: blue; ground (turned): green

Example

$P = (1, 0)$ and $\varphi = 30^\circ$ give $P' = (\cos 30^\circ, -\sin 30^\circ) = (0.866, -0.5)$.

Your turn

Turning axes

(a) $P = (1, 0)$ and $\varphi = 90°$. What are (x', y')?

(b) And with $\varphi = 180°$?

(c) A satellite is at $(26\,560, 0)$ km in the sky axes. The ground axes turn $90°$. Where is it in the ground axes?

Your turn

Turning axes

(a) $P = (1, 0)$ and $\varphi = 90°$. What are (x', y')?

(b) And with $\varphi = 180°$?

(c) A satellite is at $(26\,560, 0)$ km in the sky axes. The ground axes turn $90°$. Where is it in the ground axes?

Answers

(a) $(\cos 90°, -\sin 90°) = (0, -1)$.

(b) $(\cos 180°, -\sin 180°) = (-1, 0)$.

(c) $(0, -26\,560)$ km: the satellite "still" in the sky appears *moving west* on the ground.

In three dimensions: the order matters

On the Earth the main turn is about the z axis (the pole axis): x and y change, z **stays**.

$$\mathbf{R}_3(\varphi) = \begin{pmatrix} \cos\varphi & \sin\varphi & 0 \\ -\sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

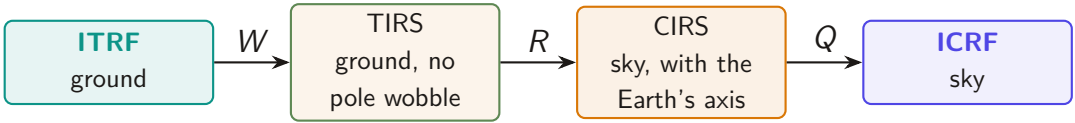
There are similar turns about x ($\mathbf{R}_1$) and y ($\mathbf{R}_2$). Any turn is a **sequence** of turns — and the sequence is a **multiplication of matrices**.

Try it with a book

Turn a book 90° about the *vertical* axis and then 90° about a *horizontal* axis. Now go back and do it **in the opposite order**.

The book ends up turned in different ways: the order of the turns matters. That is why the bridge has a fixed order.

The bridge has three steps: W, R and Q



ground $\rightarrow$ **sky:** $\vec{r}_{\mathrm{ICRF}} = Q(t)\, R(t)\, W(t)\, \vec{r}_{\mathrm{ITRF}}$ **sky** $\rightarrow$ **ground:** the opposite: turns undone (the *transposed* matrix), in reverse order.

Each step is a **small, known** matrix. What changes in time are a few angles — and those angles are the **EOP**.

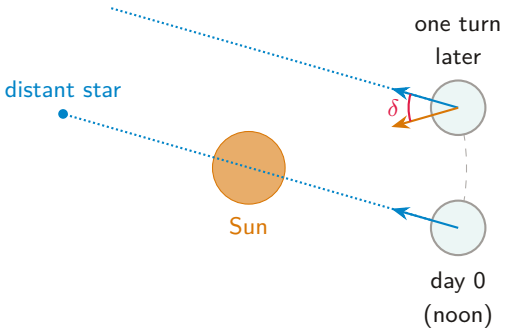
The Earth's rotation

How long is a day?

The **solar day**: the Sun returns to the same place in the sky. It lasts **24 h**.

The **sidereal day**: a star — or a quasar — returns to the same place. It lasts **23 h 56 min 4.09 s**.

Why? While it spins, the Earth also *travels* around the Sun, about 1° per day. After one full turn, the Sun is still a bit to the side: it takes about 1° more to bring it back to the same place.



Angle δ exaggerated: in reality it is $\approx 0.9856°$.

Two days, two counts

	what returns	lasts
solar day	the Sun	24 h = 86 400 s
sidereal day	a star, a quasar	86 164.09 s
difference		**3 min** 55.9 **s**

In one year there are 365.2422 solar days, but 366.2422 turns relative to the stars: the Earth makes **one more turn** than the number of days.

Test it at home

Pick a bright star behind a pole and note the time. The next day it shows up **4 minutes earlier**; a week later, $\approx$ 27 minutes earlier.

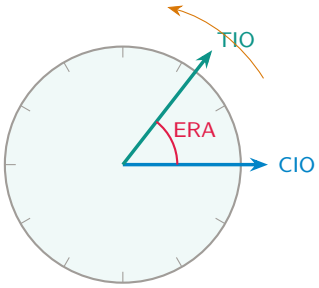
The Earth Rotation Angle (ERA)

The **ERA** (*Earth Rotation Angle*) is the "hand" that tells how far the Earth has turned relative to the still sky: the angle, about the axis, between an origin *in the sky* (the CIO) and an origin *on the ground* (the TIO).

$$\mathrm{ERA} = 360^\circ \left(0.7790572732640 + 1.00273781191135448\, T_u\right)$$

$T_u = \mathrm{JD(UT1)} - 2\,451\,545.0$ (days since J2000.0)

- on 1 Jan 2000, 12 h UT1: 280.4606°;
- grows 360.9856° per solar day;
- on 18 Oct 2026, 0 h UT1: 26.155°.



seen from above the north pole;
the Earth turns counterclockwise

Your turn

Solar day, sidereal day and the ERA

(a) A star passes behind a pole at 21:00 today. When does it pass tomorrow? And 30 days from now?

(b) How many degrees does the Earth turn in 6 h?

(c) A satellite "fixed" in the sky, at (26 560, 0) km. After 6 h, where is it in the ground axes? (use 90°)

Your turn

Solar day, sidereal day and the ERA

(a) A star passes behind a pole at 21:00 today. When does it pass tomorrow? And 30 days from now?

(b) How many degrees does the Earth turn in 6 h?

(c) A satellite "fixed" in the sky, at (26 560, 0) km. After 6 h, where is it in the ground axes? (use 90°)

Answers

(a) $\approx$ 20:56 tomorrow (-3.9 min per day); in 30 days, $30 \times 3.93 \approx 118$ min earlier: $\approx$ **19:02**.

(b) $6 \times 15.041° \approx 90.2°$.

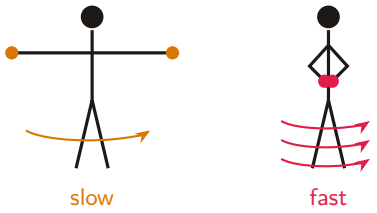
(c) (0, $-26\,560$) km: the ground point under it moved 90° west.

The Earth is not a perfect clock

A skater spins **faster** when she pulls her arms in: the total spin (the *angular momentum*) is conserved. She changed the mass distribution, and changed the pace.

The Earth does the same, on a small scale: winds, oceans, ice, the liquid core — everything that *moves* changes the pace of the spin a little.

That is why the ERA is not a calendar calculation: the time of the rotation (**UT1**) has to be **measured** every day.



$L = I\,\omega$ constant

$I \downarrow \Rightarrow\ \omega \uparrow$

What makes the rotation vary

cause	time scale	effect on the day
tides (the Moon brakes the Earth)	centuries	lengthens $\approx$ **1.8 ms per century**
winds and oceans	days to years	of the order of **1 ms** through the year
liquid core $\leftrightarrow$ mantle	decades	a few **ms**
giant earthquakes	instant	a few **μs** (Japan, 2011: ≈ 1.8 μs)

Order of magnitude

A day has 86 400 s. Variations of 1 ms are $\approx 10^{-8}$ of the day — small, but large for geodesy, as the next slide shows.

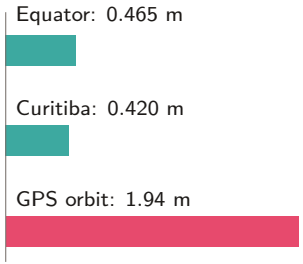
What is 1 millisecond worth?

The Earth turns $\approx 15.04''$ per second, that is, $15.04\,\mathrm{mas}$ **per millisecond**.

If the rotation clock (UT1) is off by 1 ms, it is as if the Earth were turned $15\,\mathrm{mas}$ too far or too little:

$$\text{erro} = \omega\, r\, \Delta t$$

with $\omega = 7.292115 \times 10^{-5}$ rad/s, r the distance to the axis and $\Delta t = 1$ ms.



A 1 ms error in UT1, in meters (the position is "turned" wrongly).

The axis wobbles in space

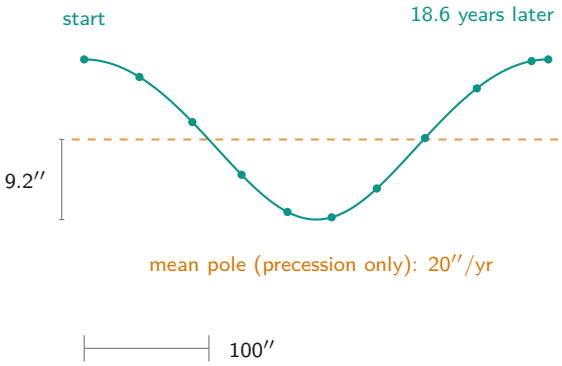
The Earth's axis, in the sky: the CIP

At each instant the Earth's axis points to a point in the sky: the **Celestial Intermediate Pole** (**CIP**).

It moves for two reasons (class 1):

- **precession**: $\approx 20''$/yr in the position of the pole;
- **nutation**: a wobble of $9.2'' \times 6.8''$, in 18.6-year cycles.

The CIP position is given by two small coordinates, X and Y, counted from the ICRS pole (in 2026, $X \approx 537''$).



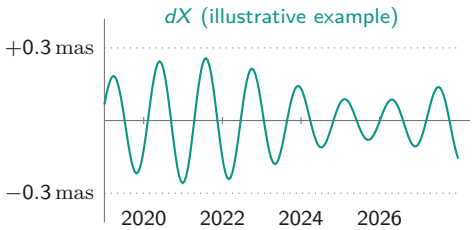
Points every 2 years. Illustrative: arbitrary phase; vertical exaggerated ≈ 7 times.

Model plus measurement: dX and dY

The IAU model (IAU 2006 precession and IAU 2000A nutation, with **1,365 terms**) computes X and Y.

But the Earth's liquid core has a natural oscillation (*free core nutation*, $\approx$ 430 days) that the model does not predict. The real pole differs from the model by up to $\approx 0.3\,\mathrm{mas}$.

VLBI measures the difference: these are **dX and dY**, published by the IERS. $0.3\,\mathrm{mas} \approx 9$ mm on the ground.



Oscillation of $\approx$ 1.2 years with variable amplitude

— so it can only be *measured*, not computed.

Your turn

The size of nutation

(a) The main nutation has an amplitude of $9.2''$. How many meters on the ground? (use $1'' = 30.9$ m)

(b) The mean pole advances $20''$ per year. In 18.6 years, how many kilometers on the ground?

(c) An error $dX = 0.3\,\mathrm{mas}$ is worth how many millimeters? (use $1\,\mathrm{mas} = 30.9$ mm)

Your turn

The size of nutation

(a) The main nutation has an amplitude of $9.2''$. How many meters on the ground? (use $1'' = 30.9$ m)

(b) The mean pole advances $20''$ per year. In 18.6 years, how many kilometers on the ground?

(c) An error $dX = 0.3\,\mathrm{mas}$ is worth how many millimeters? (use $1\,\mathrm{mas} = 30.9$ mm)

Answers

(a) $9.2 \times 30.9 \approx \mathbf{284}$ m.

(b) $20 \times 18.6 = 372'' \Rightarrow$
$372 \times 30.9 \approx 11.5$ km.

(c) $0.3 \times 30.9 \approx 9.3$ mm.

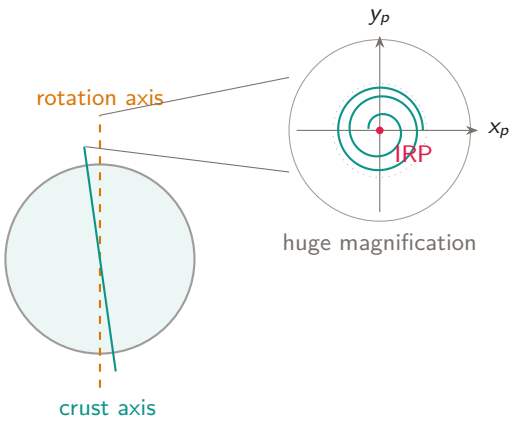
The pole moves inside the Earth

The axis also moves inside the Earth

The rotation axis is not fixed to the crust: the crust "slips" around it. The pole traces a curve of up to $\approx 0.3''$, about **9 m**, around a reference point (the **IRP**, the IERS Reference Pole).

This is the **polar motion**, described by two coordinates, x_p and y_p (toward Greenwich and toward 90° west).

It was first noticed as a small variation of the *latitude* of stations, in the late 19th century.



Not to scale: the real angle between the axes is $< 0.3''$, invisible in a drawing.

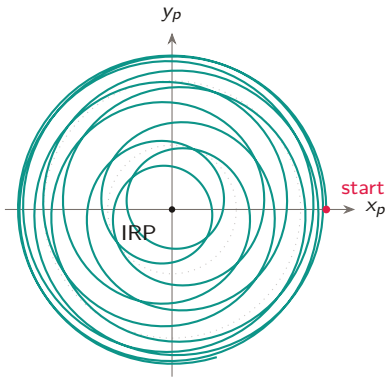
The pole's spiral

Over the years the pole draws a **spiral**: it opens and closes every ≈ 6.4 years. Between the widest and the tightest turns the radius varies from ≈ 60 to $\approx 240\,\mathrm{mas}$.

In addition, the mean pole drifts slowly, a few mas per year (centimeters per year), because of post-glacial rebound and melting ice.

How much is that?

$100\,\mathrm{mas} \approx 3.1$ m on the ground. The farthest points are ≈ 7 m from the IRP.



$100\,\mathrm{mas}$, $200\,\mathrm{mas}$: dotted circles

12.8 years (two openings). Illustrative curve from a simple model — not real data.

Two waves that add up

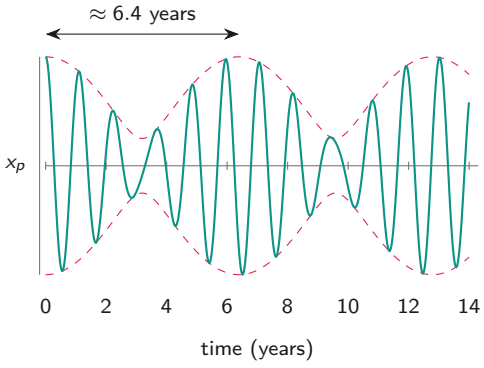
Chandler ($\approx$ 433 days): a *free* oscillation. A rigid Earth would oscillate in 305 days (Euler); since the Earth is elastic, the period lengthens. Discovered by Seth Chandler in 1891.

Annual (365 days): an oscillation *forced* by the seasons (air, water, snow moving mass).

Two waves with similar periods *beat*, like two nearly equal notes on a guitar:

$$T_{\text{bat}} = \left(\frac{1}{365.25} - \frac{1}{433} \right)^{-1}$$

$$\approx \mathbf{2\,334} \text{ days} \approx 6.4 \text{ years}$$



Sum of a 433-day wave (150 mas) and a 365-day wave (90 mas): illustrative. Dashed: the envelope.

Your turn

The size of the pole

(a) The pole moves $0.3''$. How many meters? (use $1'' = 30.9$ m)

(b) How many centimeters is $1\,\mathrm{mas}$ worth on the ground?

(c) Two waves of 365 and 433 days: how often do they come back "in phase"? (hint: $1/365 - 1/433$)

Your turn

The size of the pole

(a) The pole moves $0.3''$. How many meters? (use $1'' = 30.9$ m)

(b) How many centimeters is $1\,\mathrm{mas}$ worth on the ground?

(c) Two waves of 365 and 433 days: how often do they come back "in phase"? (hint: $1/365 - 1/433$)

Answers

(a) $0.3 \times 30.9 \approx 9.3$ m.

(b) 0.001×30.9 m ≈ 3.1 cm.

(c) $\dfrac{1}{1/365 - 1/433} \approx \mathbf{2\,300}$ days ≈ 6.3 years.

The EOP: who measures and who publishes

Five numbers per day

EOP	what it says	typical size	step
x_p, y_p	where the pole is in the crust	$\lesssim 0.3''$ ($\approx$ 9 m)	W
UT1−UTC	how far the Earth has turned, compared with the atomic clock	under 0.9 s (current rule)	R
dX, dY	correction to the axis position in the sky	$\approx$ 0.1–0.3 mas	Q

Five independent numbers. A sixth, the **LOD** (the length of day minus 86 400 s), is just the *rate* of UT1: $\mathrm{LOD} = -\,\mathrm{d}(\mathrm{UT1}-\mathrm{UTC})/\mathrm{d}t$.

These are the **EOP**: *Earth Orientation Parameters*. With them, the $Q\,R\,W$ bridge is complete.

Who measures what?

	VLBI	GNSS	SLR	DORIS
x_p, y_p	✓	✓	✓	✓
LOD	✓	✓	✓	✓
UT1−UTC	✓	rate only	rate only	rate only
dX, dY	✓	—	—	—

VLBI: antennas × quasars. GNSS: receivers × GPS and other satellites. SLR: laser on satellites with mirrors. DORIS: ground beacons × Doppler shift on satellites.

Why only VLBI?

Satellites orbit the Earth's *center of mass*: they feel the Earth below, but do not know where the "still sky" is. Only quasars do. That is why **absolute** $UT1$ and dX, dY come only from VLBI.

Who publishes?

The **IERS** (*International Earth Rotation and Reference Systems Service*, created in 1987) gathers the results of each technique's service (IVS, IGS, ILRS, IDS) and publishes the EOP.

Whoever needs the value *now* (a GNSS receiver, a VLBI antenna) uses **rapid** or **predicted** values; whoever needs the best answer waits for the **final** values.

EOP C04	final daily series, with a delay of weeks
Bulletin A	weekly: rapid values and predictions up to 1 year
Bulletin B	monthly: final values
Bulletin C	announces leap seconds (twice a year)
Bulletin D	reports DUT1 = UT1−UTC, rounded to 0.1 s

The size of each step

step	what	angle	on the ground
Q	precession since J2000 (in 2026.8)	$537''$	**16.6 km**
Q	main nutation	$9.2''$	**284 m**
W	polar motion (maximum)	$0.3''$	**9.3 m**
Q	ICRS $\times$ J2000 bias (the pole)	$17.9\,\mathrm{mas}$	**0.55 m**
R	UT1 off by 1 ms (at the Equator)	$15\,\mathrm{mas}$	**0.47 m**
Q	typical dX, dY corrections	$0.2\,\mathrm{mas}$	**6 mm**

Each step has a size. Forgetting a step gives an error of its size: that is why the bridge must be complete — and the EOP, up to date.

Try it: the GNSS simulator

In the `GNSS_simul` folder there is a globe with the GPS, GLONASS, Galileo and BeiDou satellites.

1. Turn on "Show Orbit Trajectories": the lines are the orbits *seen from the ground*.
2. GPS makes 2 turns per sidereal day: the track almost repeats every day. Galileo (17 turns in 10 sidereal days) only repeats after 10 days.
3. Click on the globe: the receiver sees the satellites above 10° and the simulator computes the DOP.

What it leaves out

It goes from sky to ground with the sidereal angle only (R). It does not apply W (pole, up to 9 m) and uses the computer clock as UT1 ($\leq$ 0.9 s: up to $\approx$ 1.7 km at the orbit). It is good for *geometry* (DOP, visibility), not for precise positioning.

Take-home — class 3

1. Still sky, turning ground: converting between them is **turning axes**. The order of the turns matters: $\vec{r}_{\mathrm{ICRF}} = Q\,R\,W\,\vec{r}_{\mathrm{ITRF}}$.
2. R**, the rotation:** the ERA angle, 360.9856° per solar day (the sidereal day lasts 23 h 56 min 4.09 s). 1 ms of error in UT1 = 0.47 m at the Equator and 1.9 m at the GPS orbit.
3. Q**, precession and nutation:** the CIP moves in the sky (20″/yr, a 9.2″ wobble); the IAU model has 1,365 terms and VLBI measures the rest (dX, $dY \approx 0.2\,\mathrm{mas}$).
4. W**, polar motion:** the pole moves up to 0.3″ (≈ 9 m) in the crust: Chandler wave (433 d) plus annual wave (365 d), beating every ≈ 6.4 years.
5. The **EOP** (x_p, y_p, UT1−UTC, dX, dY) are measured by VLBI, GNSS, SLR and DORIS — only VLBI gives absolute UT1 and dX, dY — and published by the **IERS**.

Next class: **time** — UT1, UTC, TAI, GNSS time and the leap second — and practice: from the orbit to the receiver, and Brazil in this story.

Part 2

Time and practice

class 4 of 4

Several clocks, one hour

Where we left off: class 3

In class 3:

1. the bridge between the sky and the ground has three steps: Q (precession and nutation), R (rotation) and W (polar motion);
2. five numbers per day, the **EOP**, give the angles: x_p, y_p, UT1−UTC, dX, dY;
3. 1 ms of error in UT1 = 0.47 m at the Equator and 1.9 m at the GPS orbit.

Today's question

What is UT1, where does this "rotation time" come from? And what time is it for a satellite moving at 3.9 km/s?

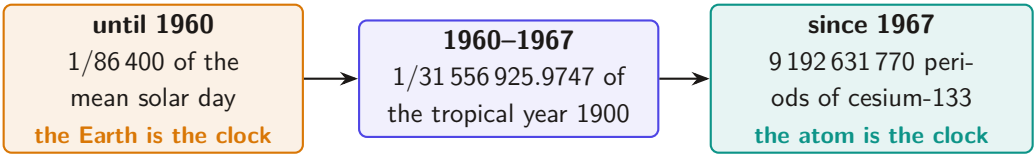
Today's questions

1. Why are there so many "times": UT1, UTC, TAI, the GPS one?
2. What is the leap second — and why did the world decide to end it, **this week**?
3. How do the satellite's orbit and the spinning Earth become a position on the ground?
4. What does this have to do with Brazil?

Deal

By the end of the class you will answer all four — and close Part 2.

The second: from the Earth to the atom



The Earth spins irregularly (class 3); the atom does **not**. That is why the second came to be defined by the atom.

How good is an atomic clock?

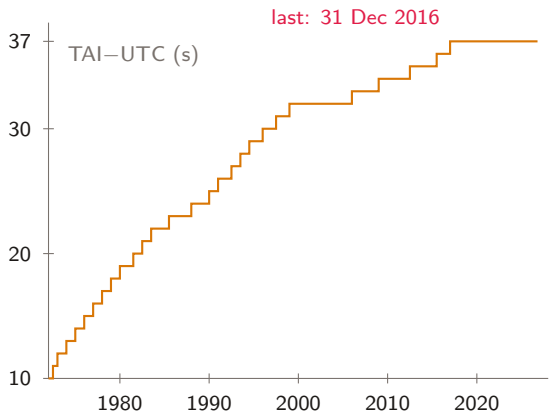
Cesium: $\approx$ 1 s in **300 million years**. Optical clocks: $\approx$ 1 s in **30 billion years** — longer than the age of the Universe (13.8 billion).

Two clocks: the Earth and the atom

	what it is	what it is for
UT1	the time of the *rotation*: comes from the ERA; irregular	orienting the Earth (the R step)
TAI	atoms: SI seconds counted without stopping, since 1958	scientific time; the basis of all the others
UTC	TAI minus a whole number of seconds; it is civil time	everyday clocks; $\lvert \mathrm{UT1} - \mathrm{UTC} \rvert < 0.9$ s (current rule)

Today: UTC $=$ TAI $-$ **37** s (since 1 Jan 2017).

The ladder of leap seconds



In 1972 UTC became TAI minus a whole number of seconds. Since then **27 leap seconds** have been inserted: 22 up to 1998 and 5 afterwards.

A day with a leap second has 86 401 s: the clock shows `23:59:60`.

In 2017 we reached TAI − UTC = 37 s. Ten years later, none more.

Why do extra seconds appear?

On average the Earth spins **more slowly** than the atomic clock: the solar day lasts slightly more than 86 400 s (tides, class 3). The difference *accumulates*.

When UT1 − UTC gets close to 0.9 s, the IERS (Bulletin C) orders a second to be inserted into UTC, at the end of June or December. UTC comes back close to UT1.

Your turn

(a) A day lasts 86 400.0015 s (1.5 ms more). In how many days does the lag add up to 0.9 s?

(b) From 1972 to 2016: 27 leap seconds in 44.5 years. One every how long?

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(b) From 1972 to 2016: 27 leap seconds in 44.5 years. One every how long?

Answers

(a) $0.9/0.0015 = \mathbf{600}$ days ≈ 1.6 yr.
(b) $44.5/27 \approx 1.6$ yr: matches!

Since 2016 the count has changed

In recent years the Earth has **sped up**: days became slightly shorter (core and mantle, class 3). And the last leap second was the one of 31 Dec 2016.

With a faster Earth, UTC would start to "fall behind": a second would have to be removed — a **negative leap second**, which has never been used.

Agnew (*Nature*, 2024): melting ice, which slows the rotation, postponed this problem by about 3 years — to $\approx$ 2029.

Climate moves the clock

Water from melting ice moves from the poles to the oceans, away from the axis: the moment of inertia *increases* and the Earth spins more slowly — like the skater opening her arms.

October 2026: the end of the leap second

On **15 Oct 2026** the 28th General Conference on Weights and Measures (CGPM, Versailles) adopted **Resolution 5**:

- UTC becomes **continuous**, with no leap seconds, from **20 May 2027**;
- the limit of $|\mathrm{UT1} - \mathrm{UTC}|$ rises to **1 hour** (3 600 s), to hold for centuries.

In 2022 the 27th CGPM (Res. 4) had already decided to enlarge that limit by 2035.

Note on rigor

The details above come from the statement of the CCTF president and the technical press, from a few days ago. The official text of Resolution 5 should be checked at the BIPM before quoting.

What this changes for geodesy

Today $|\mathrm{UT1} - \mathrm{UTC}| < 0.9$ s: ignoring the difference costs up to **419 m** on the ground (Equator) and $\approx$ **1.7 km** at the GPS orbit.

Without the jumps, UT1 $-$ UTC grows freely: from seconds to **minutes** and, only in the distant future, up to **1 hour**. Each minute is $15'$ of rotation:

$$60 \text{ s} \times 465 \text{ m/s} \approx \mathbf{28} \text{ km} \quad \text{(at the Equator)}$$

At the 1-hour limit it would be $\approx$ **1,674 km**. UTC stays TAI $-$ 37 s, with no jumps — and UT1 $-$ UTC becomes an EOP that **every program** must read from the IERS.

Who cares?

Telescopes and antennas that point using DUT1 ($\leq$ 0.9 s), planetariums, satellite software and simulators like the ones in this folder, which use the computer clock as UT1.

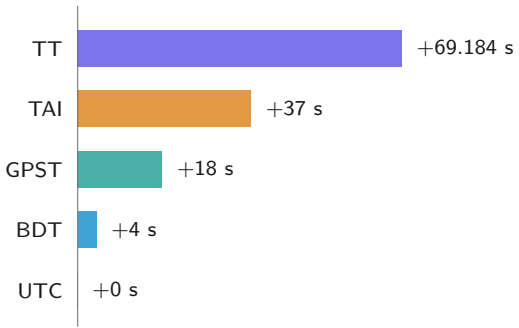
Satellite time

Each system has its own scale

system	scale	relative to UTC (2026)	leap seconds?
GPS	GPST	UTC + 18 s (= TAI − 19 s)	no
Galileo	GST	≈ GPST (tens of ns)	no
BeiDou	BDT	UTC + 4 s (= TAI − 33 s)	no
GLONASS	UTC(SU)+3 h	follows UTC (plus 3 h)	**yes**

Receivers use a leap-second parameter, sent in the navigation message, to go from GPST to UTC.

The ladder of times



Offset of each scale relative to UTC, in 2026.

TT (*Terrestrial Time*) is the time of the ephemerides: $\mathrm{TT} = \mathrm{TAI} + 32.184$ s. It enters the precession and nutation formulas (the Q step).

Rule of thumb: **UT1** for the rotation angle; **TT** for precession and nutation.

Your turn

Time sums

It is 12:00:00 UTC. What time is it in:

(a) GPST?

(b) TAI?

(c) TT?

(d) GLONASS (UTC(SU)+3 h)?

Your turn

Time sums

It is 12:00:00 UTC. What time is it in:

(a) GPST?

(b) TAI?

(c) TT?

(d) GLONASS (UTC(SU)+3 h)?

Answers

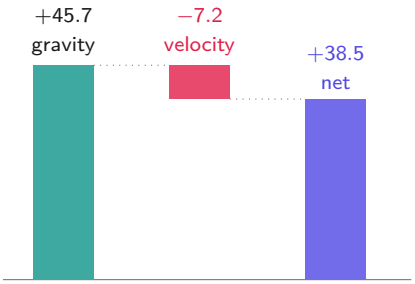
(a) 12:00:18.
(b) 12:00:37.
(c) 12:01:09.184 (37 + 32.184).
(d) 15:00:00.

Clocks in space run differently

Two Einstein effects, per day, on the GPS clock:

- up there gravity is weaker: the clock runs **faster**, $+45.7$ μs;
- the satellite moves at 3.9 km/s: the clock runs **slower**, -7.2 μs.

Balance: $+38.5$ μs per day. Uncorrected, the error would be $c \times 38.5$ μs $\approx$ **11.6 km per day**. That is why the 10.23 MHz frequency is adjusted to 10.22999999543 MHz ($\Delta f/f = 4.4645 \times 10^{-10}$) before launch.



μs per day, GPS clock relative to a clock on the ground.

Clocks that measure height

The same physics holds on the stairs of a building: a *higher* clock runs faster. The difference is tiny:

$$\frac{\Delta f}{f} = \frac{g\,h}{c^2} \approx 1.09 \times 10^{-16} \text{ per meter}$$

Optical clocks of 10^{-18} **feel 1 cm** of height difference. In 2020, two of them, at the foot and at the top of the Tokyo Skytree tower (**450 m**), measured the expected effect ($\approx 4.9 \times 10^{-14}$).

Chronometric geodesy

Comparing clocks gives the **gravitational potential** W — the same W that defines the *geoid* in the gravimetry classes. One day, clocks may replace part of leveling.

From the orbit to the receiver

The GPS orbit in two sums

In a circular orbit gravity plays the role of the centripetal force: $v^2 = GM/r$, and the period is

$$T = 2\pi\sqrt{\frac{r^3}{GM}}$$

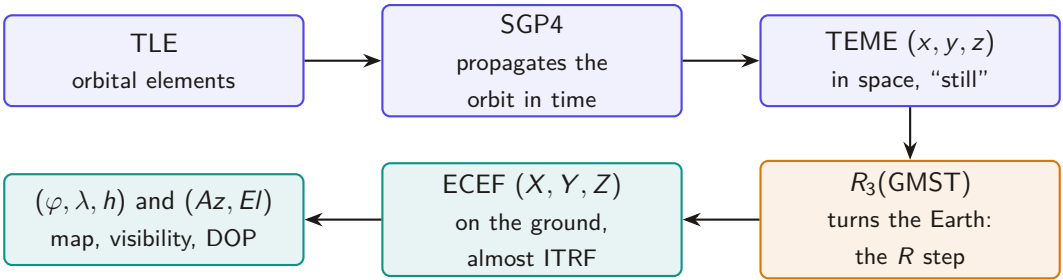
GPS makes **2 turns per sidereal day**: $T = 11$ h 58 min 2 s. Result: $r \approx 26\,562$ km (altitude $\approx 20\,184$ km), $v \approx 3.87$ km/s.

That is why the satellite's track on the ground **repeats** every sidereal day.

system	turns/days	period	altitude
GPS	2/1	11 h 58 min	20 184 km
GLONASS	17/8	11 h 16 min	19 132 km
Galileo	17/10	14 h 05 min	23 223 km
BeiDou (MEO)	13/7	12 h 53 min	21 529 km

"Turns/days": satellite turns per sidereal days until the track repeats. Nominal values, computed from the repeat ratio.

From the TLE to the map: the simulator's chain



TEME (*True Equator, Mean Equinox*): the frame in which SGP4 delivers the position. In the `GNSS_simul` (`satellite.js`) the step to the ground uses **only the sidereal angle**: no W and with UT1 = UTC.

For precision: what changes

	simulator (geometry)	precise positioning
orbit	TLE: of the order of 1 km	IGS precise orbits (cm), in the ITRF
sky → ground	R_3(GMST)	full $Q\,R\,W$, with EOP from the IERS
time	UTC in place of UT1	UT1 − UTC from the IERS; TT where due
clocks	ignored	IGS clock products + relativity
user's frame	latitude and longitude of the click	SIRGAS2000 ($\approx$ ITRF2000, epoch 2000.4)

The simulator is not wrong: it is *cheap*. It is good for seeing geometry; for centimeters, the bridge must be complete.

Your turn

The simulator's error

A simulator uses UTC in place of UT1, but today $\mathrm{UT1} - \mathrm{UTC} = -0.4$ s. Use $\omega = 7.292115 \times 10^{-5}$ rad/s.

(a) How far off is the position of a GPS satellite ($r = 26\,560$ km)?

(b) And that of a station in Curitiba (distance to the axis 5 763 km)?

(c) In which direction is the error?

Your turn

The simulator's error

A simulator uses UTC in place of UT1, but today UT1 − UTC $= -0.4$ s. Use $\omega = 7.292115 \times 10^{-5}$ rad/s.

(a) How far off is the position of a GPS satellite ($r = 26\,560$ km)?

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(c) In which direction is the error?

Answers

(a) $\omega\, r\, \Delta t = 7.292 \times 10^{-5} \times 2.656 \times 10^{7} \times 0.4 \approx$ **775** m.

(b) $7.292 \times 10^{-5} \times 5.763 \times 10^{6} \times 0.4 \approx$ **168** m.

(c) Along the parallel (*east–west*): it is a wrong turn about the axis.

Brazil and the wrap-up

Brazil in this story

The time

The **National Observatory** (Rio de Janeiro) generates the **Brazilian Legal Time** with a set of atomic clocks, linked to the BIPM's UTC (Decree 4,264/2002).

The sky

Fortaleza (CE): IVS VLBI station since 1993 (Northeast Space Radio Observatory). In Nov 2025, a new antenna with a VGOS terminal. The Southern Hemisphere has few: it helps the EOP.

The ground

The **RBMC** (IBGE) has more than 150 continuous GNSS stations: the basis of **SIRGAS2000** ($\approx$ ITRF2000, epoch 2000.4) and part of the SIRGAS-CON network.

Time, sky and ground: **the three sides of the same triangle** that this Part 2 and Part 1 drew.

Try it: the simulators' clock

In `astrometria` the Greenwich time on screen is UTC; the sky, however, turns with UT1. With $|\mathrm{DUT1}| \leq 0.9$ s the sky ends up to $13.5''$ off ($\approx$ 419 m). Precession ($\leq 0.37^\circ$ since J2000) and nutation ($9.2''$) are also missing.

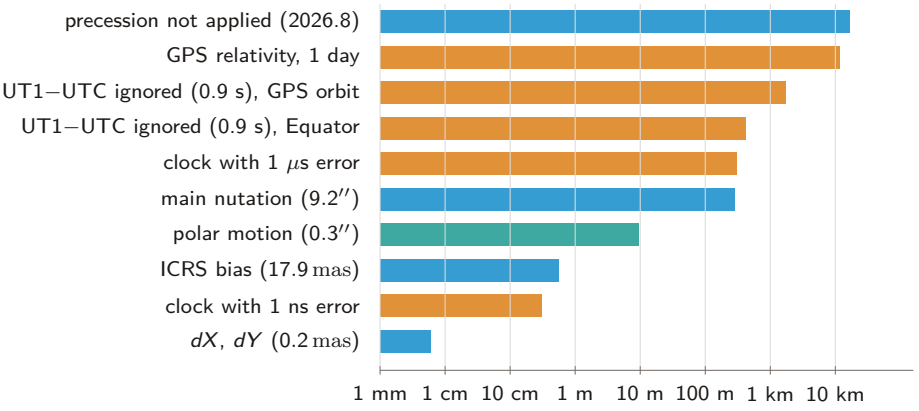
Challenges

1. Take Sirius from J2000 to 2026.8 with the formulas of appendix A4: its coordinates change by $17.3'$ ($\approx$ 32 km in an astronomical fix).
2. Replace UTC with UT1 using a UT1 $-$ UTC from the IERS and see the difference.

Order of magnitude

One arcminute in the sky is one nautical mile (1.852 km) on the ground. That is why $17.3'$ of error on the star is $\approx$ 32 km of error in the position.

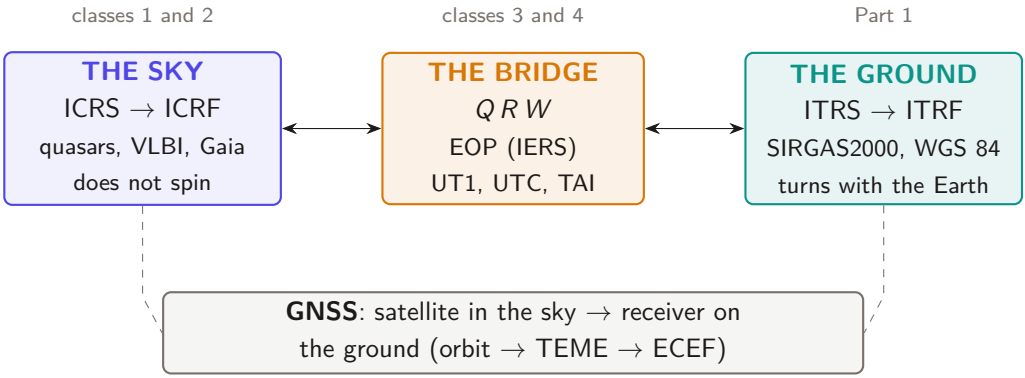
Error budget: everything at once



Effect of each step on the ground or at the orbit, on a *logarithmic* scale (each mark is 10 times the previous).

Colors: blue = Q step (sky); amber = R and time; green = W (pole).

The map of the two parts



VLBI links the two ends: antennas on the ground $\times$ quasars in the sky. The **EOP** say how the bridge is turned on each day.

Take-home — class 4 and Part 2

1. **Several times:** UT1 (rotation, irregular), TAI (atoms, continuous), UTC $=$ TAI $-$ 37 s (civil time), GPST $=$ TAI $-$ 19 s, TT $=$ TAI $+$ 32.184 s.
2. **Leap second:** 27 since 1972, the last in 2016. The 28th CGPM (15 Oct 2026) adopted continuous UTC from 20 May 2027, with a 1 h limit for $|\mathrm{UT1} - \mathrm{UTC}|$ (check the official text).
3. **GPS clocks:** $+38.5\ \mu$s/day from relativity, ≈ 11.6 km/day if uncorrected.
4. **From satellite to ground:** orbit (GPS: 2 turns per sidereal day) $\rightarrow$ TEME $\rightarrow$ $R(\mathrm{GMST}) \rightarrow$ ECEF. With precision: $Q\,R\,W$, EOP and UT1 $-$ UTC from the IERS, IGS orbits and clocks.
5. **Brazil:** ON (legal time), Fortaleza (VLBI), RBMC/SIRGAS2000 (ground).

Part 2 in one sentence: the still sky (**ICRS**/**ICRF**, quasars) and the spinning ground (ITRS/ITRF) are linked by the $Q\,R\,W$ bridge, fed by the **EOP** — measured by VLBI and satellites, published by the IERS — and by **time**.

Appendix

For those who want to go further

A1 · Normative milestones

	body	decision
1997	IAU XXIII (Kyoto), Res. B2	ICRS and ICRF (from 1 Jan 1998); Hipparcos as the optical realization
2000	IAU XXIV (Manchester), Res. B1.3, B1.6–B1.9	BCRS and GCRS; IAU 2000 precession-nutation (from 2003); CIP; CIO, TIO and ERA; redefinition of TT
2006	IAU XXVI (Prague), Res. B1	P03 precession (IAU 2006), from 1 Jan 2009
2009	IAU XXVII (Rio de Janeiro), Res. B3	ICRF2 (from 1 Jan 2010)
2018	IAU XXX (Vienna), Res. B2	ICRF3 (from 1 Jan 2019)
2021	IAU XXXI (Busan), Res. B3	Gaia-CRF3 (from 1 Jan 2022)
2022	CGPM 27, Res. 4	enlarge the $\lvert \mathrm{UT1} - \mathrm{UTC} \rvert$ limit by 2035
2026	CGPM 28, Res. 5	continuous UTC from 20 May 2027; 1 h limit (check the official text)

A2 · The QRW chain in formulas

IERS conventions (2010, ch. 5). From the ITRS to the GCRS (ground → sky):

$$[\text{GCRS}] = Q(t)\, R(t)\, W(t)\, [\text{ITRS}]$$

$$Q(t) = \begin{pmatrix} 1 - aX^2 & -aXY & X \\ -aXY & 1 - aY^2 & Y \\ -X & -Y & 1 - a(X^2 + Y^2) \end{pmatrix} \mathbf{R}_3(s), \qquad a = \frac{1}{1 + \sqrt{1 - X^2 - Y^2}}$$

$$R(t) = \mathbf{R}_3(-\text{ERA}), \qquad W(t) = \mathbf{R}_3(-s')\, \mathbf{R}_2(x_p)\, \mathbf{R}_1(y_p)$$

- X, Y: coordinates of the CIP in the GCRS (IAU 2006/2000A model plus dX, dY); s: CIO locator, $s \approx -XY/2+$ terms of μas;
- s': TIO locator, $s' \approx -47\ \mu\text{as} \times t$ (t in centuries since J2000.0);
- $\mathbf{R}_i(\alpha)$: rotation *of the axes* by α about axis i (the convention of the slide "Turning the axes").

From sky to ground: $[\text{ITRS}] = W^{\mathsf{T}} R^{\mathsf{T}} Q^{\mathsf{T}} [\text{GCRS}]$.

A3 · ERA, GMST and sidereal time

$$\mathrm{ERA} = 2\pi\,(0.7790572732640 + 1.00273781191135448\,T_u), \qquad T_u = \mathrm{JD(UT1)} - 2\,451\,545.0$$

$$\mathrm{GMST} = \mathrm{ERA} + 0.014506'' + 4\,612.156534''\,t + 1.3915817''\,t^2 - 0.00000044''\,t^3 - \ldots$$

with t in centuries of TT since J2000.0. The difference GMST − ERA is the precession in right ascension (equinox versus CIO).

on 18 Oct 2026, 0 h UT1	
ERA	26.15528°
GMST	26.49858°
GMST − ERA	1 235.9″
GMST by the simulator's formula (Meeus)	26.49860°
sidereal day by the ERA (fixed star)	86 164.0989 s
sidereal day by the GMST (equinox)	86 164.0905 s

A4 · Precession: from J2000 coordinates to today's

With $T = (\mathrm{JD(TT)} - 2\,451\,545.0)/36\,525$ and the Lieske angles (IAU 1976):

$$\zeta = 2\,306.2181'' \; T + 0.30188'' \; T^2 + 0.017998'' \; T^3$$
$$z = 2\,306.2181'' \; T + 1.09468'' \; T^2 + 0.018203'' \; T^3$$
$$\theta = 2\,004.3109'' \; T - 0.42665'' \; T^2 - 0.041833'' \; T^3$$

$$A = \cos\delta_0 \sin(\alpha_0 + \zeta), \quad B = \cos\theta \cos\delta_0 \cos(\alpha_0 + \zeta) - \sin\theta \sin\delta_0,$$
$$C = \sin\theta \cos\delta_0 \cos(\alpha_0 + \zeta) + \cos\theta \sin\delta_0, \quad \alpha = \mathrm{atan2}(A, B) + z, \quad \delta = \arcsin C$$

Example: Sirius, from J2000.0 to 18 Oct 2026

(α, δ): $(101.2872^\circ,\ -16.7161^\circ) \to (101.5865^\circ,\ -16.7457^\circ)$ by precession alone: $\Delta\alpha = +1078''$, $\Delta\delta = -106''$, separation $17.3'$. The proper motion ($1.339''$/yr) adds a further $-32.8''$ in δ.

A5 · Relativity in GNSS clocks: the sums

For a circular orbit ($v^2 = GM/a$), on-board clock against a clock on the geoid (R):

$$\frac{\Delta f}{f} = \underbrace{\frac{GM}{c^2}\left(\frac{1}{R} - \frac{1}{a}\right)}_{\text{gravity } = +5.28\times10^{-10}} - \underbrace{\frac{v^2}{2c^2}}_{\text{velocity } = -8.35\times10^{-11}} = \frac{GM}{c^2}\left(\frac{1}{R} - \frac{3}{2a}\right) \approx 4.45 \times 10^{-10}$$

In one day (86 400 s): $+45.7\ \mu$s $- 7.2\ \mu$s $= +38.5\ \mu$s. With W_0 and the Earth's rotation, $\Delta f/f = 4.4645 \times 10^{-10}$ (38.6 μs/day).

- Design frequency: $10.23\ \text{MHz}\,(1 - 4.4645 \times 10^{-10}) = 10.22999999543\ \text{MHz}$.
- Without the correction: $c \times 38.6\ \mu\text{s} \approx 11.6$ km per day.
- Eccentric orbit: periodic term $\Delta t_r = -\dfrac{2\sqrt{GM\,a}}{c^2}\, e \sin E$, about 7 ns for $e = 0.01$ (corrected by the receiver).
- On the ground: $g/c^2 = 1.0911 \times 10^{-16}$ per meter, or 1.09×10^{-18} per centimeter.

A6 · The 27 leap seconds

year	date	how many
1972	30 Jun, 31 Dec	2
1973–1979	31 Dec	7
1981–1983, 1985	30 Jun	4
1987, 1989, 1990	31 Dec	3
1992–1994	30 Jun	3
1995	31 Dec	1

year	date	how many
1997	30 Jun	1
1998	31 Dec	1
2005, 2008	31 Dec	2
2012, 2015	30 Jun	2
2016	31 Dec	1
total		**27**

Each insertion happens at `23:59:60` UTC on the date shown. TAI − UTC went from 10 s (1972) to 37 s (from 1 Jan 2017).

A7 · Glossary (1/3): systems and frames

	Portuguese	English
ICRS / ICRF	Sistema / Referencial Celeste Internacional	International Celestial Reference System / Frame
GCRS	Sistema de Referência Celeste Geocêntrico	Geocentric Celestial Reference System
ITRS / ITRF	Sistema / Referencial Terrestre Internacional	International Terrestrial Reference System / Frame
CIP	Polo Celeste Intermediário	Celestial Intermediate Pole
CIO / TIO	Origem Intermediária Celeste / Terrestre	Celestial / Terrestrial Intermediate Origin
ERA	Ângulo de Rotação da Terra	Earth Rotation Angle
GMST	Tempo Sideral Médio de Greenwich	Greenwich Mean Sidereal Time
TEME	Equador Verdadeiro, Equinócio Médio (SGP4)	True Equator, Mean Equinox (SGP4's frame)

A8 · Glossary (2/3): motions and parameters

	Portuguese	English
EOP	Parâmetros de Orientação da Terra	Earth Orientation Parameters
LOD	excesso da duração do dia sobre 86 400 s	length of day (excess over 86 400 s)
polar motion	movimento polar (x_p, y_p)	polar motion (x_p, y_p)
precession, nutation	precessão, nutação	precession, nutation
barycenter	centro de massa do Sistema Solar	center of mass of the Solar System
baseline	distância entre duas antenas	distance between two antennas
quasar	"quase uma estrela": núcleo ativo de galáxia	quasi-stellar object: active galactic nucleus
mas, μas	milissegundo e microssegundo de arco	milliarcsecond and microarcsecond

A9 · Glossary (3/3): time and services

	Portuguese	English
UT1, UTC, TAI	Tempo Universal (rotação); Coordenado; Atômico Internacional	Universal Time (rotation); Coordinated; International Atomic Time
TT	Tempo Terrestre	Terrestrial Time
GPST, GST, BDT	escalas de tempo do GPS, Galileo e BeiDou	GPS, Galileo and BeiDou system times
leap second	segundo inserido no UTC	second inserted into UTC
VLBI	Interferometria de Linha de Base Muito Longa	Very Long Baseline Interferometry
SLR, DORIS	telemetria laser a satélites; Doppler de balizas em terra	Satellite Laser Ranging; Doppler beacons (DORIS)
IERS, IVS, IGS	serviços: rotação da Terra; VLBI; GNSS	services: Earth rotation; VLBI; GNSS
BIPM, CGPM	Bureau Internacional de Pesos e Medidas; Conferência Geral	Intl. Bureau of Weights and Measures; General Conference

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- In this repository: `sgr_terrestres` (Part 1), `quase_geoide`, `gravimetria`; simulators `astrometria` and `GNSS_simul`; numbers and figures: `sgr_celestes_numeros.py`.

Values and dates checked on 18 Oct 2026. The Brazilian data and the news of the 28th CGPM came from search summaries, because some official sites were unreachable; check the primary sources before quoting.